

CHE 201 Spring 2020 HW 9

Roxanna Mendoza

TOTAL POINTS

95 / 100

QUESTION 1

1 Question 1 12.5 / 15

- 0 pts Correct
- 2.5 pts Wrong heat transfer
- 5 pts Wrong heat transfer
- 2.5 pts Wrong entropy change
- 5 pts Wrong entropy change
- ✓ - 2.5 pts Wrong T-s diagram
- 5 pts Wrong T-s diagram

QUESTION 2

2 Question 2 15 / 15

- ✓ - 0 pts Correct

QUESTION 3

3 Question 3 15 / 15

- ✓ - 0 pts Correct

QUESTION 4

4 Question 4 15 / 15

- ✓ - 0 pts Correct
- 3 pts Wrong judgement of T2
- 3 pts Wrong judgement of S2
- 3 pts Wrong judgement of S3
- 3 pts Wrong judgement of T3
- 3 pts Wrong way of COP calculation

QUESTION 5

5 Question 5 20 / 20

- ✓ - 0 pts Correct
- 2.5 pts Wrong energy balance
- 2.5 pts Wrong electronic work
- 2.5 pts Wrong heat transfer
- 2.5 pts Wrong T_b
- 2.5 pts b) Wrong Entropy balance
- 2.5 pts b) Wrong entropy production

- 2.5 pts c) Wrong Entropy balance
- 2.5 pts c) Wrong entropy production
- 1 pts keep more significant digit

QUESTION 6

6 Question 6 17.5 / 20

- 0 pts Correct
- 2.5 pts Wrong final temperature
- 5 pts Wrong final temperature
- 2.5 pts Wrong final pressure
- 5 pts Wrong final pressure
- 2.5 pts Wrong entropy balance
- 5 pts Wrong entropy balance
- ✓ - 2.5 pts Wrong entropy calculation
- 5 pts Wrong entropy calculation

Homework #9

Roxanna Mendoza

#1.

$\frac{1}{10}$ kmol of carbon Monoxide (CO)

$$P_1 = 150 \text{ kPa}, T_1 = 300 \text{ K}$$

$$P_2 = 500 \text{ kPa}, T_2 = 370 \text{ K}$$

$$W = -300 \text{ kJ}$$

$$@ T_1 = 300 \text{ K}$$

$$u = 6,229 \text{ kJ/k.mol}$$

$$s = 197.723 \text{ kJ/k.mol.K}$$

$$@ T_2 = 370 \text{ K}$$

$$u = 7,689 \text{ kJ/k.mol}$$

$$s = 203.942 \text{ kJ/k.mol.K}$$

Part A:

$$\Delta U = m(u_2 - u_1)$$

$$\Delta U = (\frac{1}{10})(7,689 - 6,229)$$

$$\Delta U = 146 \text{ kJ}$$

$$Q = \Delta U + W$$

$$Q = (146 \text{ kJ}) + (-300 \text{ kJ})$$

$$Q = -154 \text{ kJ}$$

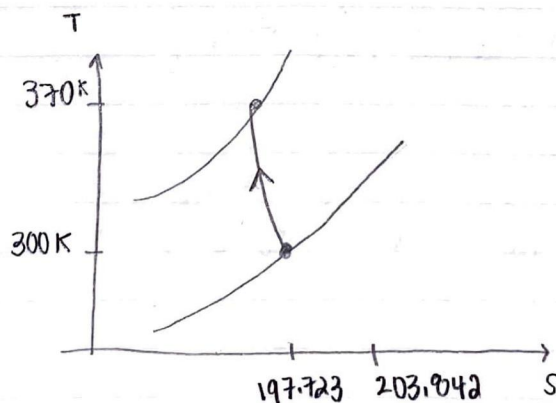
Part B:

$$S_2 - S_1 = S^o(T_2) - S^o(T_1) - R \ln \left(\frac{P_2}{P_1} \right)$$

$$\Delta S = (\frac{1}{10}) [203.942 - 197.723] - 0.314 \ln \left(\frac{500}{150} \right)$$

$$\Delta S = -0.389 \text{ kJ/K}$$

Diagram:



1 Question 1 12.5 / 15

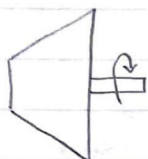
- 0 pts Correct
- 2.5 pts Wrong heat transfer
- 5 pts Wrong heat transfer
- 2.5 pts Wrong entropy change
- 5 pts Wrong entropy change
- ✓ - 2.5 pts Wrong T-s diagram
- 5 pts Wrong T-s diagram

#2.

Steam enters turbine operating at steady state; $\Delta KE = \Delta PE = 0$

$$P_1 = 1 \text{ MPa}, T_1 = 200^\circ\text{C}$$

$$T_2 = 40^\circ\text{C}, X = 83\% = 0.83$$



① closed system

② steady state

③ $\Delta KE = \Delta PE = 0$ ④ $\dot{Q} = 0$ Energy Balance: $\dot{W} = \dot{m}(h_1 - h_2)$; $\frac{\dot{W}}{\dot{m}}$ ← power developed

$$\textcircled{a} T_1 = 200^\circ\text{C} \text{ \& } P_1 = 1 \text{ MPa}; h_1 = 2827.9, S_1 = 6.6940 \quad \text{TABLE A-4}$$

$$\textcircled{c} T_2 = 40^\circ\text{C}; h_{2f} = 167.57; h_{2fg} = 2406.7, S_{2f} = 0.5725, S_{2g} = 8.2570$$

$$h_2 = h_{2f} + X h_{2fg}$$

$$h_2 = 167.57 + (0.83)(2406.7) = 2165.131 \text{ kJ/kg}$$

$$S_2 = S_{2f} + X(S_{2g} - S_{2f})$$

$$S_2 = 0.5725 + (0.83)(8.2570 - 0.5725) = 6.9906 \text{ kJ/kg}\cdot\text{K}$$

Part A:

$$\frac{\dot{W}}{\dot{m}} = (h_1 - h_2)$$

$$\frac{\dot{W}}{\dot{m}} = (2827.9 - 2165.131) = 662.769 \text{ kJ/kg}$$

Part B:

$$\Delta S = S_2 - S_1$$

$$\Delta S = 6.9906 - 6.6940 = 0.2966 \text{ kJ/kg}\cdot\text{K}$$

2 Question 2 15 / 15

✓ - 0 pts Correct

#3. $P_1 = 10 \text{ lbf/in}^2$, $T_1 = 500^\circ\text{F}$; $h_1 = 1287.7$, $u_1 = 1182.2$, $S_1 = 1.969$
 $P_2 = 80 \text{ lbf/in}^2$, $T_2 = 800^\circ\text{F}$; $h_2 = 1430.4$, $u_2 = 1292.4$, $S_2 = 1.867$

° $\Delta PE = \Delta KE = 0$

° closed system

$$Q = \int_1^2 T ds \Rightarrow \frac{Q}{m} = \int_1^2 T ds$$

$$\frac{Q}{m} = T_{\text{avg}} (S_2 - S_1)$$

$$T_{\text{avg}} = \frac{1}{2} (T_1 + T_2)$$

$$500^\circ\text{F} = 959.67^\circ\text{R}$$

$$800^\circ\text{F} = 1259.67^\circ\text{R}$$

$$\frac{Q}{m} = 0.5 (1259.67 + 959.67) (1.867 - 1.969)$$

$$Q/m = -109.86 \text{ Btu/lb}$$

$$\Delta u = \frac{Q}{m} - \frac{W}{m}$$

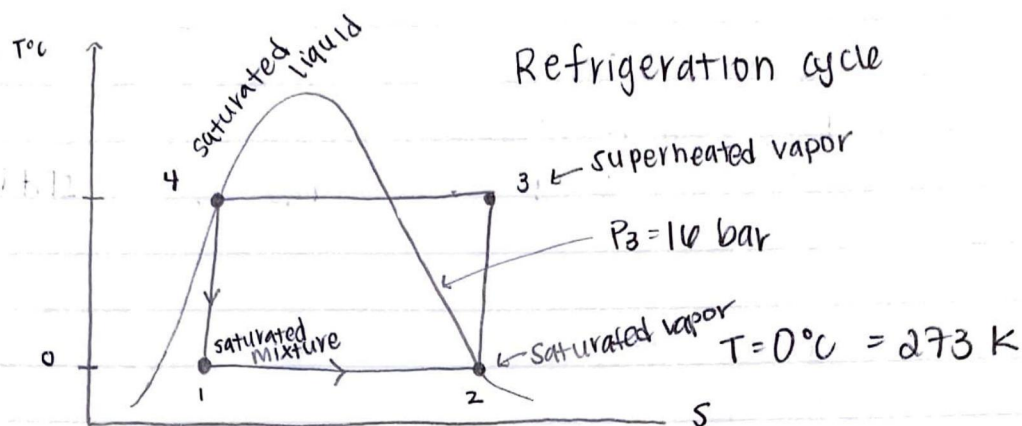
$$(1292.4 - 1182.2) = (-109.86) - \frac{W}{m}$$

$$\frac{W}{m} = 220.06 \text{ Btu/lb}$$

3 Question 3 15 / 15

✓ - 0 pts Correct

#4.



@ $T = 0^\circ\text{C}$ $S_g = 0.9190 \text{ kJ/kg}\cdot\text{K}$

$S_2 = S_3 = 0.9190$

At $P = 16 \text{ bar}$ $S_3 = 0.9190$ therefore interpolation
Between $T = 60^\circ\text{C}$ and $T = 70^\circ\text{C}$:

$$\frac{T_3 - 60^\circ\text{C}}{70^\circ\text{C} - 60^\circ\text{C}} = \frac{0.9190 - 0.9069}{0.9457 - 0.9069}$$

$T_3 = 63.12^\circ\text{C} \Rightarrow 336.27 \text{ K}$

Coefficient of Performance

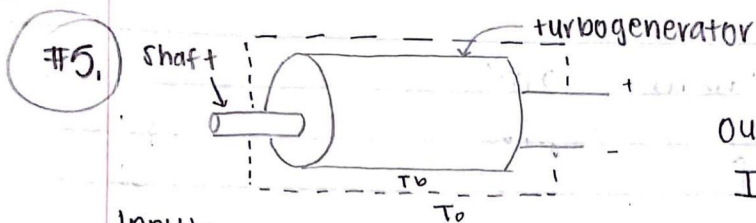
$$(COP)_{\text{ref}} = \frac{T_c}{T_h - T_c}$$

$COP = \frac{273.15}{336.27 - 273.15} = 4.33$

4 Question 4 15 / 15

✓ - 0 pts Correct

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- 3 pts Wrong judgement of T3
- 3 pts Wrong way of COP calculation



Output:

$$I = 230 \text{ Amp ; Voltage} = 13,000 \text{ V}$$

Input:

$$\omega = 1800 \text{ RPM ; Torque} = 16,700 \text{ N}\cdot\text{m}$$

$$\frac{dE}{dt} = \dot{Q} - W \quad \begin{array}{l} \text{output} \\ \text{input} \end{array}$$

$$0 = \dot{Q} - (W_{\text{electricity}} - W_{\text{shaft}})$$

$$0 = \dot{Q} - (I \cdot V - T \cdot \omega)$$

$$\omega = \frac{1800 \text{ REV}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ sec}} \cdot \frac{2\pi \text{ rad}}{1 \text{ REV}} = 188.5 \frac{\text{rad}}{\text{s}}$$

$$0 = \dot{Q} - (230 \cdot 13,000 - 16,700 \cdot 188.5)$$

$$\dot{Q} = -157,950 \text{ W} \Rightarrow \dot{Q} = -157.95 \text{ kW}$$

Part A:

$$\dot{Q} = -hA(T_b - T_o) \text{ where } h = 110 \text{ W/m}^2\cdot\text{K}, A = 32 \text{ m}^2, T_o = 298 \text{ K}$$

$$-157,950 = - (110) \cdot (32 \text{ m}^2) \cdot (T_b - 298)$$

$$-157,950 = -3520 T_b + 1048960$$

$$T_b = 342.87 \text{ K}$$

Part B:

→ steady state $\therefore$ equal to zero

$$\frac{dS}{dt} = \frac{\dot{Q}}{T_b} + \dot{S}$$

$$0 = \frac{-157.95 \text{ kW}}{342.87 \text{ K}} + \dot{S} \Rightarrow \dot{S} = 0.461 \text{ kW/K}$$

Part C:

$$0 = \frac{-157.95 \text{ kW}}{298 \text{ K}} + \dot{S}$$

$$\dot{S} = 0.530 \text{ kW/K}$$

5 Question 5 20 / 20

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- 1 pts keep more significant digit

#6. $V_1 = 1.0 \text{ m}^3$
 $T_1 = 400 \text{ K}$
 $P_1 = 3 \text{ bar}$



$V_{2 \text{ initial}} = 1.0 \text{ m}^3$
 $T_{2 \text{ initial}} = 400 \text{ K}$
 $P_{2 \text{ initial}} = 1.5 \text{ bar}$

Part A:

$\Delta U = Q - W \quad \Delta U = 0$

$\Delta U_1 + \Delta U_2 + \Delta U_{\text{piston}} = 0$

$m_1 [u(T_f) - u(400 \text{ K})] + m_2 [u(T_f) - u(400 \text{ K})] = 0$

$u(T_f) = u(400 \text{ K})$

$T_f = 400 \text{ K}$

Part B: Total volume = 2 m^3

$P_f = \frac{(m_1 + m_2) R T_f}{V} ; m_1 = \frac{P_1 (V_{12})}{R T_1} , m_2 = \frac{P_2 (V_{12})}{R T_2}$

$P_f = \left(\frac{P_1 (V_{12})}{R T} \right) + \left(\frac{P_2 (V_{12})}{R T} \right) \frac{R T}{V}$

$P_f = \frac{P_1 + P_2}{2} \Rightarrow P_f = \frac{3 + 1.5}{2} \Rightarrow P_f = 2.25 \text{ bar}$

Part C:

$\Delta S = \int_1^2 \left(\frac{\delta Q}{T} \right)_b + \sigma \quad \text{OR} \quad \sigma = \Delta S \Rightarrow \sigma = \Delta S_1 + \Delta S_2 + \Delta S_{\text{piston}}$

$\Delta S_1 = m_1 \left[S^\circ(T_f) - S^\circ(T_1) - R \ln \frac{P_f}{P_1} \right] = -R m_1 \ln \left(\frac{P_f}{P_1} \right)$

$\Delta S_2 = m_2 \left[S^\circ(T_f) - S^\circ(T_2) - R \ln \frac{P_f}{P_2} \right] = -R m_2 \ln \left(\frac{P_f}{P_2} \right)$

$\therefore \sigma = -R \left[m_1 \ln \left(\frac{P_f}{P_1} \right) + m_2 \ln \left(\frac{P_f}{P_2} \right) \right] = -R \left[\frac{P_1 (V_{12})}{R T_1} \ln \left(\frac{P_f}{P_1} \right) + \frac{P_2 (V_{12})}{R T_2} \ln \left(\frac{P_f}{P_2} \right) \right]$

$= -\frac{V_{12}}{T} \left[P_1 \ln \left(\frac{P_f}{P_1} \right) + P_2 \ln \left(\frac{P_f}{P_2} \right) \right]$

$= -\frac{1}{400} \left(3 \times \ln \left(\frac{2.25}{3} \right) + 1.5 \ln \left(\frac{2.25}{1.5} \right) \right) \cdot \left(\frac{100 \text{ kJ}}{1 \text{ bar}} \right) \left(\frac{1 \text{ kJ}}{1 \text{ kN} \cdot \text{m}} \right) = -0.2426 \frac{\text{kJ}}{\text{K}}$

3 bar

6 Question 6 17.5 / 20

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